

FRÉDÉRIC MAGOULÈS

BSc, MSc, PhD, FIMA, FBCS

Professor of Applied Mathematics and Scientific Computing

CURRICULUM VITAE

EDUCATION

- Habilitation to Direct Research (Habilitation à Diriger des Recherches), Applied Mathematics, Université Pierre et Marie Curie, France (2005)
- PhD, Applied Mathematics, Université Pierre et Marie Curie, France (2000)
- MSc, Numerical Analysis, Université Pierre et Marie Curie, France (1995)
- MSc, Applied Mathematics, Université Pierre et Marie Curie, France (1994)
- BSc, Applied Mathematics, Université Pierre et Marie Curie, France (1993)

ACADEMIC APPOINTMENTS

- **Transvalor S.A. Academic Chair**, Co-Inaugural Holder, CentraleSupélec, France (2022-2026)
- **Professor**, Laboratoire de Mathématiques et Informatique pour la Complexité et les Systèmes, CentraleSupélec, Université Paris-Saclay, France (2015-present)
- **Director**, Centre of Integrated Engineering for Sustainable Society, Groupe des Écoles Centrales, Châtenay-Malabry, France & Yokohama, Japan (2010-2014)
- **SAP Academic Chair**, Inaugural Holder, École Centrale Paris, France (2008-2013)
- **Professor**, Laboratoire de Mathématiques Appliquées aux Systèmes, École Centrale Paris, Châtenay-Malabry, France (2006-2015)
- **Associate Professor**, Institut Elie Cartan de Nancy, Université Henri Poincaré, Nancy, France (2005-2006)
- **Assistant Professor**, Institut Elie Cartan de Nancy, Université Henri Poincaré, Nancy, France (2000-2005)
- **Temporary Assistant Professor**, Laboratoire Jacques-Louis Lions, Université Pierre et Marie Curie, Paris, France (1999-2000)

AWARDS, HONOURS, AND RECOGNITION

Fellowships in Professional Societies

- Fellow of the British Computer Society, United Kingdom (2015)
- Fellow of the Institute of Mathematics and its Applications, United Kingdom (2014)

Major Awards

- IBM Champion, IBM, USA (2018)
- NVIDIA GPU Teaching Center Award, NVIDIA, USA (2014)
- Microsoft Research Academic Award, Microsoft, USA (2014)
- NVIDIA GPU Research Center Award, NVIDIA, USA (2013, 2014)
- IBM Faculty Award, IBM, USA (2012)

Who's Who Listings

- Who's Who in Computational Science and Engineering (2005-2006)

HONORARY AND VISITING ACADEMIC TITLES

- **Visiting Professor**, School of Computing and Mathematical Sciences, University of Greenwich, London, United Kingdom (2014-2017, 2023-2026 and 2026-2029)
- **Distinguished Guest Professor**, School of Computer and Information Engineering, Hubei University, China (2019-present)
- **Chair Professor**, Department of Computer Science and Communication Engineering, Providence University, Taichung City, Taiwan (2017-2024; seven consecutive annual appointments, from 2017-2018 to 2023-2024).
- **Distinguished Chair Professor**, Department of Information Engineering and Computer Science, Feng Chia University, Taichung City, Taiwan (2016-2017)
- **Visiting Professor**, Department of Computer Science and Communication Engineering, Providence University, Taichung City, Taiwan (2016)
- **Adjunct Professor**, Laboratoire d'Informatique de l'École Polytechnique, École Polytechnique, Palaiseau, France (2015-2016)
- **Visiting Professor**, School of Mathematics and Computer Sciences, Jiangnan University, Wuhan, China (2014-present)
- **Guest Professor**, School of Computer Science and Technology, Wuhan University of Science and Technology, Wuhan, China (2014-2017)
- **Visiting Professor**, Energy Conversion Research Center, Doshisha University, Kyotanabe-shi, Japan (2014-2019; five consecutive annual appointments, from 2014-2015 to 2018-2019).
- **Honorary Professor**, Faculty of Engineering and Information Technology, University of Pécs, Pécs, Hungary (2012-present)
- **Visiting Fellow**, Energy Conversion Research Center, Doshisha University, Kyotanabe-shi, Japan (2011-2012, 2012-2013 and 2013-2014).

SHORT-TERM ACADEMIC VISITS

- Visiting Professor, École Centrale de Pékin, Beihang University, Beijing, China, Oct.4-16 (2018)
- Visiting Professor, Department of Architecture and Civil Engineering, Peoples' Friendship University of Russia, Moscow, Russia, May.6-12 (2018)
- Visiting Professor, École Centrale de Pékin, Beihang University, Beijing, China, Oct.8-20 (2016)
- Visiting Professor, Wuhan University of Technology, Wuhan, China, Nov.21-29 (2014)
- Visiting Professor, Department of Mechanical Engineering, Kyushu University, Nishi-ku, Fukuoka, Japan, Oct.3-14 (2012)
- Visiting Professor, Barcelona Supercomputing Center, Barcelona, Spain, Apr.25-May 21 (2012)
- Visiting Professor, Department of Mechanical and Industrial Engineering, Concordia University, Montréal, Canada, Sep.19-26 & Nov.22-30 (2011)
- Visiting Professor, Department of Computer Sciences and Information Mathematics, University of Electro-Communications, Chofu-shi, Tokyo, Japan, Sep.22-Oct.2 (2007)
- Visiting Professor, Department of Mathematics, Keio University, Hiyoshi Kohoku-ku, Tokyo, Japan, Jan.10-23 (2007)
- Visiting Professor, Department of Civil Engineering, Chuo University, Bunkyo-ku, Tokyo, Japan, Jan.8-16 (2006)
- Visiting Professor, School of Computing and Mathematical Sciences, University of Greenwich, London, United Kingdom, Nov.1-18 & Dec.1-18 (2005)
- Visiting Professor, Department of Civil Engineering, Chuo University, Bunkyo-ku, Tokyo, Japan, May 9-30 (2005)
- Visiting Professor, School of Engineering and Physical Sciences, Heriot Watt University, Riccarton, Scotland, United Kingdom, May 1-30 (2004)
- Visiting Professor, Department of Mechanical Sciences and Engineering, Tokyo Institute of Technology, Meguro-ku, Tokyo, Japan, Mar.1-27 (2004)
- Visiting Professor, School of Engineering and Physical Sciences, Heriot Watt University, Riccarton, Scotland, United Kingdom, Nov.24-Dec.21 (2002)
- Visiting Professor, Edinburgh Parallel Computing Center, University of Edinburgh, Edinburgh, Scotland, United Kingdom, Apr.22-Jun.17 (2002)
- Visiting Scientist, Department of Civil Engineering, Chuo University, Bunkyo-ku, Tokyo, Japan, Nov.13-27 (1999)
- Visiting Scientist, Department of Computer Sciences and Information Mathematics, University of Electro-Communications, Chofu-shi, Tokyo, Japan, Oct.16-30 (1999)
- Visiting Scientist, Edinburgh Parallel Computing Center, University of Edinburgh, Edinburgh, Scotland, United Kingdom, Jan.11-Apr.19 (1999)
- Visiting Scientist, Department of Aerospace Engineering Sciences, University of Colorado, Boulder, Colorado, USA, Oct.1-Dec.1 (1998)
- Visiting Scientist, Department of Aerospace Engineering Sciences, University of Colorado, Boulder, Colorado, USA, Oct.19-Dec.17 (1997)

PROFESSIONAL SERVICE ACTIVITIES

Professional Societies

- Fellow, The Institute of Mathematics and its Applications (IMA), UK
- Fellow, The British Computer Society (BCS), UK
- Member, Association Française pour l'Intelligence Artificielle (AFIA), France (2025-present)
- Member, The Association for Computing Machinery (ACM), US (2014-present)
- Member, The British Computer Society (BCS), UK (2014-present)
- Member, The Institute of Mathematics and its Applications (IMA), UK (2014-2015)
- Member, The Society for Industrial and Applied Mathematics (SIAM), US (2013-present)
- Member, The Institute of Electrical and Electronics Engineers (IEEE), US (2013-2018)
- Member, Société de Mathématiques Appliquées et Industrielles (SMAI), France (2001-2011)

Editor-in-Chief, Editor and Editorial Boards

Extensive editorial experience with leading international publishers, including roles as Editor-in-Chief, Series Editor, Associate Editor, and Editorial Board member for numerous high-impact journals and book series in applied mathematics, computational science, and engineering.

Active contributions to the scientific publishing process over more than two decades.

(Complete list below)

- Series Editor, Wiley-ISTE Science Publishing Ltd Series on *Computer Engineering*, published by **Wiley-ISTE** (2015-present)
- Member of the Board of Editors, *Numerical Algorithms*, published by **Springer** (2022-present)
- Associate Editor, *Frontiers of Computer Science*, published by **Springer** (2012-2022)
- Member of the Board of Editors, *Advances in Engineering Software*, published by **Elsevier** (2009-present)
- Member of the Board of Editors, *Engineering Applications of Computational Fluid Mechanics*, published by **Taylor & Francis**, Florida, USA (2006-present)
- Associate Editor and Europe Representative, *Journal of Algorithms and Computational Technology*, published by SAGE (2006-present)
- Editor-in-Chief, Chapman & Hall, CRC Press' Series on *Numerical Analysis and Scientific Computing*, published by **Taylor & Francis**, Florida, USA (2005-present)
- Member of the Board of Editors, *International Journal of Computer Mathematics*, published by **Taylor & Francis**, Florida, USA (2005-2011)
- Member of the Board of Editors, *Pollack Periodica: An International Journal for Engineering and Informatics*, published by Akadémiai Kiadó, Budapest, Hungary (2006-present)
- Member of the Board of Editors, *Acta Technica Jaurinensis*, published by Széchenyi István Univ., Hungary (2018-present)
- Member of the Board of Editors, *Structural Mech. of Eng. Constructions and Buildings*, published by Peoples' Friendship Univ. of Russia, Russia (2018-present)

Guest-Editor of Journal Special Issues

Served as guest editor for international journals, contributing to the development of special issues in applied mathematics and computational science.

(Selected list below)

- Guest Editor of the Special Issue Embedded multicore computing and applications, in *The Journal of Supercomputing* (2017)
- Guest Editor of the Special Issue Advances in High Performance Computing: on the path to Exascale software, in *Advances in Engineering Software* (2017)
- Guest Editor of the Special Issue Embedded multicore computing and applications, in *Concurrency and Computation: Practice and Experience* (2016)
- Guest Editor of the Special Issue Innovative Algorithms for Extreme Scale Computing, in *International Journal of High Performance Computing Applications* (2015)
- Guest Editor of the Special Issue in Honor of Professor Hideo Kawarada on the Occasion of his 70th Birthday, in *Journal of Computational and Applied Mathematics* (2009)
- Guest Editor of the Special Issue Techniques of Scientific Computing in Engineering, in *Journal of Algorithms and Computational Technologies* (2009)
- Guest Editor of the Special Issue Innovative Algorithms in Science and Engineering, in *International Journal of Computer Mathematics* (2008)
- Guest Editor of the Special Issue Distributed Algorithms in Science and Engineering, in *International Journal of Computer Mathematics* (2007)
- Guest Editor of the Special Issue Domain Decomposition Methods: Recent Advances and new Challenges in Engineering, in *Computer Methods in Applied Mechanics and Engineering* (2007)
- Guest Editor of the Special Issue High Performance Computing in Science and Engineering, in *Advances in Engineering Software* (2007)
- Guest Editor of the Special Issue High Performance Computing for Computational Mechanics, in *Computers and Structures* (2007)
- Guest Editor of the Special Issue Techniques of Scientific Computing for Energy and Environment, in *International Journal of Energy, Environment and Economics* (2006)
- Guest Editor of the Special Issue Parallel and Distributed Computing for Computational Mechanics, in *Applied Mathematical Modelling* (2006)
- Guest Editor of the Special Issue Parallel and Vector Processing in Science and Engineering, in *Applied Mathematical Modelling* (2006)
- Guest Editor of the Special Issue High Performance Computing for Wave Propagation, in *Journal of Computational Acoustics* (2006)
- Guest Editor of the Special Issue Absorbing Boundary Conditions, in *Computer Methods in Applied Mechanics and Engineering* (2006)
- Guest Editor of the Special Issue Innovative Computational Methods for Wave Propagation, in *Journal of Computational Acoustics* (2005)

Advisory Board and Committees

Serving as a committee member and/or referee for the following international publishers, association and agencies:

- **Government Agencies :** Agence Nationale de la Recherche (ANR, France), Agence Nationale Recherche Technologie (ANRT, France), Comisión Nacional de Investigación (CONICYT, Chile), Institut Farman (École Normale Supérieure de Cachan, France), Fonds de recherche sur la société et la culture (FRQSC, Canada), FEMTO (Université de Franch Comté, France), Fonds de la Recherche Scientifique (F.R.S.-FNRS, Belgium), National Research Foundation (NRF, Singapore), Research Grants Council (GRC, Hong Kong) Partnership for Advanced Computing in Europe (PRACE, Europe), The Leverhulme Trust (London, UK), Croatian Science Foundation (CR, Croatia), Czech Science Foundation (GACR, Czech Republic),
- **Research Excellence Framework :** Haut conseil de l'évaluation de la recherche et de l'enseignement supérieur (HCERES, France),
- **National Agencies :** Certificat de Compétences en Calcul Intensif (C3I), CPU (Conférences des Présidents d'Universités), GENCI (Grand Equipement National pour le Calcul Intensif) et la Maison de la Simulation,
- **Publishers :** Dunod (France), Chapman & Hall / CRC Press (USA, UK, Singapore)
- **Journals :** Advances in Eng. Software, Applied Acoustics, Applied Energy, Applied Math. Modelling, Applied Mathematics and Computation, Applied Numerical Mathematics, Computers and Fluids, Comp. Meth. in Applied Mech. and Eng., Comp. & Structures, Concurrent Engineering: Research and Applications, Energy, Energy Efficiency, IEEE TNNLS, IEEE PES Transactions on Power Systems, Int. J. of AdHoc and Ubiquitous Computing, Int. J. of Distributed Sensor Networks, Int. J. of Energy, Environment and Economics, Int. J. for Numerical and Analytical Meth. in Geomechanics, Int. J. for Numerical Meth. in Eng., J. of Algorithms and Computation Technology, J. of Comp. Acoustics, J. of Computational and Applied Mathematics, J. of Computational Science, J. of Comp. Physics, J. of Sound and Vibration, Mathematical Problems in Engineering, M2AN, Numerical Algorithms, Parallel Computing, Pollack Periodica, Review Renewable & Sustainable Energy Reviews, Scientific World Journal, SIAM J. of Numerical Analysis, SIAM J. of Sc. Computing, Software Practice & Experience, Tehnicki Vjesnik, and other international journals.

Workshop and Conference Committees

Extensive international involvement in the organization of scientific conferences and workshops, with participation in more than 90 events worldwide. Roles include General Chair, Program Chair, Workshop Chair, Steering Committee member, and Scientific Committee member across major conferences in computational science, high-performance computing, and applied mathematics.

(Selected roles and complete list below)

- Member of the Conference Editorial Boards. Thirteenth International Conference on Engineering Computational Technology, Budapest, Hungary, August 27-29, (2026).
- General Chair. SMAI-SeMA XXII Jacques-Louis Lions Hispano-French School on Numerical Simulation in Physics and Engineering, Paris, January 12-15 (2026).
- Member of the Scientific Committee. 22nd International Miklos Ivanyi PhD & DLA Symposium, Pécs, Hungary, October (2025).
- Member of the Scientific Committee. 21th International Miklos Ivanyi PhD & DLA Symposium, Pécs, Hungary, October (2024).
- General Chair (with Barry Topping, Peter Ivanyi). The Seventh International Conference on Parallel, Distributed, GPU and Cloud Computing for Engineering. Pécs, Hungary, August 28-31st (2023).

- Member of the Scientific Committee. 20th International Miklos Ivanyi PhD & DLA Symposium, Pécs, Hungary, October (2023).
- Member of the Steering Committee and Member of the Scientific Committee. 20th Conference on Distributed Computing and Applications for Business Engineering and Science, Nanning Normal University, Guangxi, China, December 10-12 (2021).
- Workshop Chair. Deep Learning for Computational Science Workshop, Nvidia Deep Learning Institute, September 7-8 (2021) (online event).
- Workshop Chair. Fundamentals of Deep Learning Workshop, Nvidia Deep Learning Institute, December 15 (2020) (online event).
- Member of the Scientific Committee and Chairman of one session. 16th International Miklos Ivanyi PhD & DLA Symposium, Pécs, Hungary, October 26-27 (2020) (online event).
- Member of the Steering Committee and Member of the Scientific Committee. 19th International Symposium on Distributed Computing and Applications for Business Engineering and Science, Xuzhou, China, October 16-19 (2020).
- Workshop Chair. Fundamentals of Accelerated Computing Workshop, Nvidia Deep Learning Institute, September 7 (2020) (online event).
- Member of the International Advisory Committee. 3rd International Conference on Computational Engineering and Science for Safety and Environmental Problems, Kobe, Japan, March 8-11 (2020).
- Co-Workshop Chair (with Che-Lun Hung). 1st International Workshop on High Performance Computing and Applications in conjunction with 16th International Symposium on Pervasive Systems, Algorithms and Networks, Naples, Italy, September 16-20 (2019).
- Member of the Program Committee. 13th International Conference on Advanced Engineering Computing and Applications in Sciences, Porto, Portugal, September 22-26 (2019).
- Member of the Steering Committee and Member of the Program Committee. 18th Distributed Computing and Applications in Business, Engineering, and Sciences Conference, Hubei University, Wuhan, China, November 8-10 (2019).
- Member of the Program Committee. 15th International Symposium on Pervasive Systems, Algorithms, and Networks, Italy, October (2019).
- Member of the Scientific Committee. Mathematical Modelling and Computational Methods in Applied Sciences and Engineering, Olomouc, Czech Republic, September 16-20 (2019).
- Member of the Conference Editorial Board. 16th International Conference on Civil, Structural and Environmental Engineering Computing, Lake Garda, Italy, September 16-19 (2019).
- Co-Organizer (with Christian Glusa) of a Minisymposium on Asynchronous Iterative Methods. International Congress on Industrial and Applied Mathematics, Valencia, Spain, July 15-19 (2019).
- Member of the Conference Editorial board, Organizer of a Minisymposium on GPU Computing for Engineering Analysis, Simulation and Design and Co-Organizer (with Pierre Spiteri) of a Minisymposium on Asynchronism in Parallel Computation. 6th International Conference on Parallel, Distributed, GPU and Cloud Computing for Engineering, Pécs, Hungary, June 4-5 (2019).
- Member of the Scientific Committee. 1st University Mohammed VI Polytechnic Spring School on Complexity Analysis of Industrial Systems and Advanced Modeling Ben Guerir, Morocco, April 22-27 (2019).
- Member of the Technical Program Committee. 12th International Conference on Advanced Engineering Computing and Applications in Sciences, Athens, Greece, November 18-22 (2018).
- Member of the Scientific Committee and Chairman of one session. 14th Miklós Iványi International PhD-DLA Symposium, Pécs, Hungary, October 29-30 (2018).

- Member of the Program Committee. 17th International Symposium on Distributed Computing and Applications for Business Engineering and Science, Wuxi, China, October 19-23 (2018).
- Member of the Conference Editorial Board. 10th International Conference on Engineering Computational Technology, Sitges, Barcelona, Spain, September 4-6 (2018).
- Workshop Chair. Paris-Metagenomic Analysis Group, Gif-sur-Yvette, France, February 2 (2018).
- Member of the Scientific and Advisory Committee. 2nd International Conference on Modern Mathematical Methods and High Performance Computing in Science & Technology, Inderprastha Engineering College Ghaziabad, UP India, January 4-6 (2018).
- Member of the Scientific Committee and Chairman of one session. 13th Miklós Iványi International PhD-DLA Symposium, Pécs, Hungary, November 3-4 (2017).
- Member of the Program Committee. 16th International Symposium on Distributed Computing and Applications for Business Engineering and Science, AnYang, China, October 13-16 (2017).
- Co-Workshop Chair (with Che-Lun Hung). 2nd International Workshop on Innovative Computational Technologies and Its Applications in conjunction with 11th International Conference on Frontier of Computer Science and Technology, Exeter, Devon, UK, June 21-23 (2017).
- Member of the Program Committee. 14th International Symposium on Pervasive Systems, Algorithms, and Networks, Exeter, Devon, UK, June 21-23 (2017).
- Member of the Scientific Committee. 5th International Conference on Parallel, Distributed, Grid and Cloud Computing for Engineering, Pécs, Hungary, May 30-31 (2017).
- General Chair. 1st International Conference on Computational Science, Paris, France, November 28-December 2 (2016).
- Co-Workshop Chair (with Che-Lun Hung). 6th International Workshop on Embedded Multi-core Computing and Applications at 1st International Conference on Computational Science, Paris, France, November 28-December 2 (2016).
- Member of the Scientific Committee and Chairman of one session. 12th Miklós Iványi International PhD-DLA Symposium, Pécs, Hungary, 3rd-4th November (2016).
- Member of the Program Committee. Workshop on Computer Networks, Web Service/Technologies, and Software Defined Networking at the International Computer Symposium, National Chung Cheng University, Taiwan, December 15-17 (2016).
- Co-General Chair (With Julien Bourgeois). 19th IEEE International Conference on Computational Science and Engineering, Paris, France, August 24-26 (2016).
- Co-General Chair (With Julien Bourgeois). 14th IEEE International Conference on Embedded and Ubiquitous Computing, Paris, France, August 24-26 (2016).
- Co-General Chair (With Julien Bourgeois). 15th International Symposium on Distributed Computing and Applications for Business, Engineering, and Sciences, Paris, France, August 24-26 (2016).
- Co-Workshop Chair (with Che-Lun Hung). 6th International Workshop on Embedded Multi-core Computing and Applications at 19th IEEE International Conference on Computational Science and Engineering, Paris, France, August 24-26 (2016).
- Workshop Chair. 4th Workshop on Analytical and Numerical Methods for Scientific Computing in Science and Engineering, Paris, France, August 24-26 (2016).
- Member of the Program Committee. 7th International Conference on Future Computational Technologies and Applications, Rome, Italy, March 20-24 (2016).
- Member of the Scientific/Advisory Committee. International conference on Modern Mathematical Methods and High Performance Computing in Science & Technology, Ghaziabad, Uttar Pradesh, India, December 26-31 (2015).

- Member of the Organizing Committee. 6th International Conference on Complex Systems Design & Management, Paris, France, November 23-25 (2015).
- Co-Workshop Chair (with Che-Lun Hung). International Workshop on Multicore Computing at 14th IEEE International Conference on Ubiquitous Computing and Communications, Liverpool, UK, October 26-28 (2015).
- Member of the International Scientific Committee. 11th Miklós Iványi International PhD-DLA Symposium in Pécs, Hungary, October 19-20 (2015).
- Member of the Conference Editorial Board and Member of the Best Paper Competition Committee. 15th International Conference on Civil, Structural and Environmental Engineering Computing, Prague, Czech Republic, September 1-4 (2015).
- Member of the Program Committee. 17th IEEE International Conference on High Performance Computing and Communications, New York, U.S.A., August 24-26 (2015).
- Co-Workshop Chair (with Che-Lun Hung). 5th International Workshop on Embedded Multi-core Computing and Applications at 17th IEEE International Conference on High Performance Computing and Communications, New York, USA, August 24-26 (2015).
- Member of the Program Committee. 14th International Symposium on Distributed Computing and Applications for Business Engineering and Science, Guiyang, China, Aug. 18-24 (2015).
- Member of the Conference Editorial Board and Member of the Best Paper Competition Committee. 4th International Conference on Parallel, Distributed, Grid and Cloud Computing for Engineering, Dubrovnik, Croatia, March 24-27 (2015).
- Member of the Program Committee. 6th IEEE International Conference on cloud Computing technology and Science, Singapore, December 15-18 (2014).
- Member of the Program Committee and Chairman of two sessions. 13th International Symposium on Distributed Computing and Applications to Business, Engineering & Science, XianNing, China, November 24-27 (2014).
- Member of the International Scientific Committee. 10th International PhD-DLA Symposium, Pécs, Hungary, October 20th-21st (2014).
- Member of the Conference Editorial Board and Member of the Best Paper Competition Committee. 9th International Conference on Engineering Computational Technology, Naples, Italy, September 2-5 (2014).
- Co-General Chair (with Julien Bourgeois). 16th IEEE International Conference on High Performance Computing and Communications, Paris, France, August 20-22 (2014).
- Co-General Chair (with Julien Bourgeois). 6th IEEE International Symposium on Cyberspace Safety and Security, Paris, France, August 20-22 (2014).
- Co-General Chair (with Julien Bourgeois). 11th IEEE International Conference on Embedded Software and Systems, Paris, France, August 20-22 (2014).
- Member of the Program Committee. Intelligent Computing and Math Model Conference, Wuhan, China, June 13-16 (2014).
- Member of the International Advisory Committee. 1st International Conference on Computational Engineering and Science for Safety and Environmental Problems, Sendai International Center, Sendai, Japan, April 13-16 (2014).
- Member of the Program Committee. 16th IEEE International Conference on Computational Science and Engineering, Sydney, Australia, December 3-5 (2013).
- Program Vice Chair. 15th IEEE International Conference on High Performance Computing and Communications, Zhangjiajie, China, November 13-15 (2013).

- Member of the Steering Committee and Member of the Scientific Committee. 1st International Workshop on Context for Business Process Management, Annecy, Haute-Savoie, France, October 28-29 (2013).
- Member of the International Scientific Committee and Chairman of one session. 9th International PhD-DLA Symposium, Pécs, Hungary, October 21-22 (2013).
- Member of the Program Committee and Chairman of one session. 12th International Symposium on Distributed Computing and Applications to Business, Engineering & Science, United Kingdom, September 2-4 (2013).
- Member of the Conference Editorial Board and Member of the Best Paper Competition Committee. 3rd International Conference on Parallel, Distributed, Grid and Cloud Computing for Engineering, Pécs, Hungary, March 25-27 (2013).
- Workshop Chair. 3rd Workshop on Analytical and Numerical Methods for Scientific Computing in Science and Engineering, Châtenay Malabry, France, Feb.28-Mar.1 (2013).
- Member of the Program Committee. 15th IEEE International Conference on Computational Science and Engineering, Paphos, Cyprus, December 5-7 (2012).
- Member of the Scientific Committee and Chairman of two sessions. 8th International PhD-DLA Symposium, Pécs, Hungary, October 20-30 (2012).
- Member of the Program Committee. 11th International Symposium on Distributed Computing and Applications to Business, Engineering & Science, Guilin, China, October 19-22 (2012).
- Co-Organizer (with Hiroshi Kanayama, Daisuke Tagami, Shuyu Sun) of a Minisymposium on Recent Progress in Computational Fluid Dynamics and Related Topics (four sessions) and Chairman of one session. International Computational Mechanics Symposium, Japan Society of Mechanical Engineers, Kobe, Japan, October 9-11 (2012).
- Member of the Conference Editorial Board and Member of the Best Paper Competition Committee. 8th International Conference on Engineering Computational Technology, Dubrovnik, Croatia September 4-7 (2012).
- Chairman of one session. 21st International Conference on Domain Decomposition Methods, Inria Rennes-Bretagne-Atlantique, France, on June 25-29 (2012).
- Member of the Scientific Committee. 7th International PhD-DLA Symposium, Pécs, Hungary, October 24-25 (2011).
- Member of the Paper Committee. 10th International Symposium on Distributed Computing and Applications to Business, Engineering & Science, Wuxi, China, October 14-17 (2011).
- Member of the Conference Editorial Board. 2nd International Conference on Parallel, Distributed and Grid Computing for Engineering, Ajaccio, Corsica, France, April 12-15 (2011).
- Member of the Conference Editorial Board. 7th International Conference on Engineering Computational Technology, Valencia, Spain, September 14-17 (2010).
- Member of the Program Committee. 9th International Symposium on Distributed Computing and Applications To Business, Engineering & Science, Hong Kong, China, August 10-12 (2010).
- Workshop Chair. 2nd Workshop on Analytical and Numerical Methods for Scientific Computing in Science and Engineering, Châtenay Malabry, France, February 25-26 (2010).
- Co-Workshop Chair (with Frédéric Abergel, Erick Herbin) and Chairman of one session. 10ème Rencontre SMAI Mathématiciens Industriels: Evaluation des risques, Châtenay Malabry, France, May 14 (2009).
- Member of the Conference Editorial Board. 1st International Conference on Parallel, Distributed and Grid Computing for Engineering, Pécs, Hungary, April 6-8 (2009).

- Member of the Conference Editorial Board. 6th International Conference on Engineering Computational Technology, Athens, Greece September 2-5 (2008).
- Co-Workshop Chair (with Takashi Kako). International Workshop on Applied Mathematics and Computational Science, Chofu-shi, Japan, September 25-28 (2007).
- Chairman of one session. 2nd International Conference on Grid and Pervasive Computing, Paris, France, May 2-6 (2007).
- Member of the Conference Editorial Board. 5th International Conference on Engineering Computational Technology, Las Palmas de Gran Canaria, Spain, September 12-15 (2006).
- Workshop Chair. 1st Workshop on Analytical and Numerical Methods for Scientific Computing in Science and Engineering, Nancy, France (2006).
- Member of the Paper Committee, Co-Organizer (with Choi-Hong Lai) of a Minisymposium on Science and Engineering Applications (four sessions) and Chairman of two sessions. Distributed Computing and Applications to Business, Engineering & Science, Greenwich, United Kingdom, August 25-27 (2005).
- Member of the Conference Editorial Board. 4th International Conference on Engineering Computational Technology, Lisbon, Portugal, September 7-9 (2004).
- Co-Organizer (with Takashi Kako) of a Minisymposium on Domain Decomposition Methods in Acoustics and Related Topics II (three sessions) and Chairman of two sessions. 13th International Conference on Domain Decomposition Methods, Lyon, France, October 9-12 (2000).
- Co-Organizer (with Takashi Kako) of a Minisymposium on Domain Decomposition Methods in Acoustics and Related Topics I (two sessions) and Chairman of one session. 12th International Conference on Domain Decomposition Methods, Chiba, Japan, October 25-29 (1999).
- Organizer of a Minisymposium on Domain Decomposition Methods for the Helmholtz Problem and Chairman of one session and Panellist in a discussion panel. 4th International Conference on Theoretical and Computational Acoustics, Trieste, Italy, May 10-14 (1999).
- Member of the Scientific Committee, Organizer of a Minisymposium on Parallel Algorithms and Computational Acoustics (two sessions) and Chairman of two sessions. 3rd Euro-Conference on Parallel and Distributed Computing for Computational Mechanics, Weimar, Germany, March 20-25 (1999).

PhD and HDR Committees

Extensive involvement in doctoral and habilitation committees, with participation in more than 55 PhD defenses and several HDR juries (excluding candidates under my supervision or from my research group), serving as reviewer, examiner, and president.

(Complete list below)

1. Member of the PhD thesis committee of Kevin Kana (Université Marie et Louis Pasteur, France, 2025)
2. Member of the PhD thesis committee of Darío Martínez Martínez (Universidad de Castilla-La Mancha, Spain, 2024)
3. Member (President) of the PhD thesis committee of Juan Carlos Martinez Rocha (Université de Versailles, France, 2024)
4. Member (President) of the PhD thesis committee of Helena Shourick (Université de Lyon, France, 2023)

5. Member (President) of the PhD thesis committee of Maksim Berezov (Université Paris Sciences & Lettres, 2022)
6. Member of the PhD thesis committee of Thapelo Cornelius Mosetlhe (Tshwane University of Technology, South Africa, 2021)
7. Member of the PhD thesis committee of Mehdi Hafsia (Université de Versailles Saint Quentin en Yvelines, France, 2020)
8. Reviewer of the PhD thesis of ChengXin Yin (Université d'Orléans, France, 2020)
9. Reviewer of the PhD thesis of Bruno Derias (Université de Toulouse, France, 2020)
10. Reviewer (and President) of the PhD thesis of Aref Harakeh (Université d'Aix-Marseille, France, 2019)
11. Reviewer of the PhD thesis of Etienne Gay (Université Paris-Diderot, France, 2019)
12. Member (President) of the PhD thesis committee of Florian Gouin (MINES ParisTech, France, 2019)
13. Member (President) of the PhD thesis committee of Maxime Martelli (Univ. Paris Sud, France, 2019)
14. Member of the PhD thesis committee of Amjad Mouhaidali (Université Claude Bernard Lyon 1, France, 2019)
15. Reviewer of the PhD thesis of Pierre-Marie Gibert (Université Claude Bernard Lyon 1, France, 2018)
16. Member (President) of the PhD thesis committee of Alexis Zima (Université Sorbonne Paris Cité, France, 2018)
17. Reviewer of the PhD thesis of Vincent Parimbene (Institut National Polytechnique de Toulouse, France, 2018)
18. Reviewer of the PhD thesis of Mohammed Bakiri (Université Bourgogne Franche-Comté, France, 2018)
19. Reviewer of the PhD thesis of Bashar Talib Al-Nuaimi (Université Bourgogne Franche-Comté, France, 2017)
20. Reviewer of the PhD thesis of Camille Negrello (École Normale Supérieure de Paris-Saclay, France, 2017)
21. Member (President) of the PhD thesis committee of Paul Oumaziz (École Normale Supérieure Paris-Saclay, France, 2017)
22. Member of the PhD thesis committee of Imad Kissami (Université Paris 13, Villetaneuse, France, 2017)
23. Reviewer of the HDR thesis of David Horák (Technical University of Ostrava, Czech Republic, 2017)
24. Reviewer of the PhD thesis of Thibault Notargiacomo (Université Grenoble Alpes, France, 2017)
25. Member (President) of the PhD thesis committee of Hugo Martin (Université de Versailles Saint Quentin en Yvelines, France, 2016)
26. Reviewer of the PhD thesis of Charles Emile Ramamonjisoa (Université de Franche-Comté, France, 2016)
27. Member (President) of the PhD thesis committee of Ian Masliah (Université Paris Sud, France, 2016)
28. Reviewer of the PhD thesis of Subodh Paudel (École des Mines de Nantes, France, 2016)
29. Reviewer of the PhD thesis of Quan Huu Do (Université de Franche-Comté, France, 2016)

30. Reviewer of the PhD thesis of Sylvain Mercier (Université de Toulouse, France, 2015)
31. Reviewer of the PhD thesis of Petros Aristidou (University of Liege, Belgium, 2015)
32. Reviewer of the HDR thesis of Cheuk Ming Chau (Institut National Polytechnique de Toulouse, France, 2015)
33. Reviewer of the PhD thesis of Vincent Reverdy (Observatoire de Paris, France, 2014)
34. Reviewer of the PhD thesis of Ange Toulougoussou (Université Pierre et Marie Curie, France, 2014)
35. Reviewer of the PhD thesis of Lambert Trénoras (Université de Versailles Saint Quentin en Yvelines, France, 2014)
36. Reviewer of the PhD thesis of Laurent Berenguer (Université Claude Bernard, France, 2014)
37. Reviewer of the PhD thesis of Sari Lakkis (Université de Versailles Saint Quentin en Yvelines, France, 2014)
38. Reviewer of the PhD thesis of Mata Khalili (Tohoku University, Japan, 2014)
39. Member (President) of the PhD thesis committee of Frédéric Goncalves (Université de Versailles Saint Quentin en Yvelines, France, 2014)
40. Member (President) of the PhD thesis committee of Thiam Chekhou (Université de Toulouse, France, 2014)
41. Member of the PhD thesis committee of Ane Beatriz Eguzkitza Bazar (Universitat Politècnica de Catalunya, Barcelona, 2014)
42. Reviewer of the PhD thesis of Lilia Ziane Khodja (Université de Franche-Comté, France, 2013)
43. Reviewer of the PhD thesis of Anima Gacem (Université de Versailles Saint Quentin en Yvelines, France, 2013)
44. Reviewer of the PhD thesis of Matthieu Durut (Télécom ParisTech, France, 2012)
45. Member (President) of the PhD thesis committee of Sébastien Demmel (Queensland University of Technology, Australia, 2012)
46. Member (President) of the PhD thesis committee of Amine Arezki (Université de Versailles Saint Quentin en Yvelines, France, 2012)
47. Member of the PhD thesis committee of Rishaad Abdoola (Tshwane University of Technology, Republic of South Africa, 2012)
48. Reviewer of the PhD thesis of Pierre-Yves Aquilanti (Université Lille 1 Sciences et Technologies, France, 2011)
49. Reviewer of the PhD thesis of The Tung Nguyen (Institut National Polytechnique de Toulouse, France, 2011)
50. Reviewer of the PhD thesis of Antonio López Márquez (University of Almería, Spain, 2010)
51. Reviewer of the PhD thesis of Shahin Ghomeshi (Concordia University, Canada, 2010)
52. Reviewer of the PhD thesis of Olfa Hamdi-Larbi (Université de Tunis El Manar, Tunis, 2010)
53. Reviewer of the PhD thesis of Malek Baklouti (Université de Versailles Saint Quentin en Yvelines, France, 2009)
54. Member of the PhD thesis committee of Zhang Ye (Université des Sciences et Technologies de Lille, France, 2009)
55. Member of the PhD thesis committee of Ronan Paugam (École Centrale Paris, France, 2008)
56. Reviewer of the PhD thesis of Raul Baños Navarro (Universidad de Almeria, Spain, 2006)
57. Reviewer of the PhD thesis of Hugo Fol (Université de Nice, France, 2006)
58. Reviewer of the PhD thesis of Faten Jelassi (ENITunis, Tunisie, 2006)
59. Member of the PhD thesis committee of Olga Egorova (Tokyo Institute of Technology, Japan, 2004)

KNOWLEDGE DISSEMINATION

- **Media Coverage:** Research featured in major media outlets, including Capital, Le Nouvel Observateur, Le Point, L'Express, Challenges, Enjeux Les Echos, L'Expansion, L'Usine Nouvelle, etc.
- **Science Communication:** Authored popular science articles aimed at disseminating research to a broad audience.

KEYNOTE AND INVITED DISTINGUISHED LECTURES

- Twenty-first Seminar on Numerical Analysis, Ostrava, Czech Republic, Jan.25-29 (2021)
- Seventh European Seminar on Computing, Pilsen, Bohemia, Czech Republic, Jun.8-12 (2020)
- Journée Scientifique FEMTO, Besançon, France, Jun.26 (2018)
- Fifth International Conference on Parallel, Distributed, Grid and Cloud Computing for Engineering, Pécs, Hungary, May 30-31 (2017)
- Journée High Performance Computing and Domain Decomposition Methods, Villetaneuse, France, Apr. (2016)
- Fifth IMACS Conference on Mathematical Modelling and Computational Methods in Applied Sciences and Engineering, in honour of Professor Owe Axelsson's 80th birthday, Roznov pod Radhostem, Czech Republic, Jun.2-6 (2014)
- Journée Aristote sur l'équation du millénaire, Palaiseau, France, Mar.12 (2014)
- Inaugural Lecture at Maison de la simulation de Champagne-Ardenne, Reims, France, Dec.17 (2013)
- Twelfth International Symposium on Distributed Computing and Applications to Business, Engineering & Science, London, United Kingdom, Sep.2-4 (2013)
- Third International Conference on Parallel, Distributed, Grid and Cloud Computing for Engineering, Pécs, Hungary, Mar.25-27 (2013)
- Inaugural Lecture at Eighth International PhD & DLA Symposium, Pécs, Hungary, Oct. 29-30 (2012)
- International Computational Mechanics Symposium, in honour of the Division of Computational Mechanics's 25th anniversary, Japan Society of Mechanical Engineering, Kobe, Japan, Oct.9-11 (2012)
- Seventh International Conference on Engineering Computational Technology, Dubrovnik, Croatia, Sep.4-7 (2012)
- Journée Aristote sur le calcul hybride, Palaiseau, France, Mar.25 (2010)
- International Workshop on Applied Mathematics and Computational Science, in honour of Professor Hideo Kawarada's 70th birthday Chofu-shi, Japan, Sep.25-28 (2007)
- Japan Society of Mechanical Engineering (JSME) Workshop, Tokyo, Japan, Mar.22 (2004)
- Japan Society for the Promotion of Sciences (JSPS) Workshop, Chofu-shi, Japan, Mar.22 (2004)
- Journée parallélisme en mécanique des solides et des structures, Paris, France (2000)
- Third Euro-Conference on Parallel and Distributed Computing for Computational Mechanics, Weimar, Germany, Mar.20-25 (1999)

COLLOQUIA, SEMINARS, AND INVITED TALKS

Delivered over 120 talks between 1996 and 2026 at national and international institutions, including leading universities in Europe, North America, and Asia.

Topics covered Numerical Analysis, High Performance Computing, and Scientific Machine Learning.

(Detailed list available upon request)

SHORT COURSES TAUGHT

- “Asynchronous Iterative Methods”, XXI Jacques-Louis Lions Hispano-French School on Numerical Simulation in Physics and Engineering, University of Castilla-La Mancha, Ciudad Real, Spain (Summer 2025)
- “Introduction to Digital Twins in Artificial Intelligence”, CentraleSupélec Executive Education, Paris, France (Spring and Autumn 2024), (Spring and Autumn 2025)
- “Introduction to High Performance Computing in Civil Engineering”, Peoples’ Friendship University of Russia, Moscow, Russia, (Autumn 2021)
- “Asynchronous Iterative Methods”, Winter school Modelling and Simulation of Challenging Engineering Problems, Ostrava, Czech Republic, (Winter 2021)
- “Parallel Computations in Structural Mechanics”, Peoples’ Friendship University of Russia, Moscow, Russia (Spring 2018)
- “High Performance Computing”, École Centrale Pékin, Beijing, China (Autumn 2016 & 2018)
- “Asynchronous Domain Decomposition Methods and Parallel Processing for Acoustic Scattering”, Wuhan University of Science and Technology, Wuhan, China (Autumn 2014)
- “Advanced Asynchronous Computational Methods for Computational Fluid Dynamics”, Doshisha University, Kyotanabe-shi, Japan (Spring 2014)
- “Advanced Computational Methods for Power Systems”, Summer School, Electric Energy Systems-University Enterprise Training Partnership, University Duisburg-Essen, Duisburg, Germany (Summer 2012)
- “High Performance Computing”, Summer School, École Centrale Paris/Executive Education, Châtenay Malabry, France (Summer 2007, 2008, 2009, 2010, 2011)
- “Domain Decomposition Methods and Parallel Processing for Acoustic Scattering”, University of Electro-Communications, Chofu-shi, Japan (Autumn 2007)
- “Introduction to Parallel Scientific Computations”, Chuo University, Tokyo, Japan (Autumn 2005)
- “Parallelising Large Structures and Acoustics Codes”, LMS International, Leuven, Belgium (Autumn 1998)
- “Domain Decomposition and Parallel Processing in Structural Mechanics”, University of Electro-Communications, Chofu-shi, Japan (Autumn 1997)

TEACHING AND TRAINING

Université Paris-Saclay

Courses Taught at CentraleSupélec (formerly École Centrale Paris)

- Theoretical and Computational Acoustics for Pollution Analysis - Graduate level.
- Advanced Theoretical and Numerical Methods in Acoustic Pollution Control - Graduate level.
- High-Performance Computing for Carbon Footprint Reduction - Graduate level.
- Parallel and Distributed Computing - Graduate level.
- Numerical Methods for Computational Fluid Dynamics - Graduate level.
- Design and Programming of Parallel Algorithms - Graduate level.
- Advanced Parallel and Distributed Algorithms - Graduate level.
- High-Performance Computing in Engineering and Computational Finance - Graduate level.
- Introduction to High-Performance Computing - Graduate level.
- Numerical Methods for Coupled Field Problems - Graduate level.
- Signal and Image Processing with MATLAB - Graduate level.
- Introduction to MATLAB - Graduate level.
- Advanced Programming and Software Development - Graduate level.
- Programming and Algorithm Design with Python. - Graduate level.

Courses Taught at École Polytechnique

- Parallel and Distributed Computing - Graduate level.

Courses Taught at École Supérieure des Sciences Economiques et Commerciales

- Advanced Linear Algebra - Undergraduate level.
- Linear Algebra for Engineering - Undergraduate level.
- Algorithmic and Software Concepts - Undergraduate level.
- Programming for Data Science - Undergraduate level.

Lecture Notes

- [1] F. Magoulès and F.-X. Roux. Calcul scientifique parallèle: Cours, exemples avec OpenMP et MPI, exercices corrigés. Collection Cours de CentraleSupélec, Dunod, Paris, France. Paperback: 252 pages, 2ème edition, Oct 2020.
- [2] F. Magoulès and F.-X. Roux. Calcul scientifique parallèle: Cours, exemples avec OpenMP et MPI, exercices corrigés. Collection Cours de l'Ecole Centrale Paris, Dunod, Paris, France. Paperback: 288 pages, Mar 2013.

Doctoral Students Supervised

- Marien Chenaud, Hybrid Scientific Machine Learning Methods for Surrogate Modeling of Physical Systems (2023-2026)
- Xuewen Xiao, Automation of Database Generation, Statistical Analysis, and Supervised Training of Graph Neural Networks: Application to the Spectroscopy of Large CH Clusters of Astrophysical Interest (2022-2025) - Research Engineer at China National Petroleum Corporation, China.

- Zixing Qiu, Efficient implementation of Deep Learning techniques for the spectroscopic identification of molecules using Tensor Decomposition methods (2022-2025).
- Ahmed El Kerim, Asynchronous domain decomposition method in structural mechanics (2019-2023) - Research Engineer at Safran Tech, France.
- Mengchen Wang, Asynchronous simulation approach for interactive simulation in fluid mechanics (2017-2021) - Research Engineer at MTB by Créative (MTB111), France.
- Qinneng Zou, Iterative methods with retards for the solution of large-scale linear systems (2016-2019) - Assistant Professor at Beijing University of Posts and Telecommunications, China.
- Florian Plaza Oñate, Abundance-based reconstitution of microbial pan-genomes from whole-metagenome shotgun sequencing data: Application to the study the human gut microbiota (2014-2018) - Research Engineer at INRA, France.
- José Garay, Asynchronous Optimized Schwarz Methods for Partial Differential Equations in Rectangular Domains (2015-2018) - Post-Doctoral Fellow at Louisiana State University, USA.
- Guillaume Benissan, Asynchronous domain decomposition methods for massively parallel computing (2014-2017) - Assistant Professor at Peoples' Friendship University of Russia, Russia.
- Hanyu Zhang, Retarded Iterative Methods for Parallel Computations (2013-2016) - Assistant Professor at Providence University, Taiwan.
- Ge Song, Parallel and Continuous Join Processing for Data Streams (2013-2016) - Research and Development Engineer at INRIA Startup ActiveEon, France.
- Abal-Kassim Cheik Ahamed, Parallel Iterative Methods on Graphics Processing Units (2012-2015) - Assistant Professor at Université de Mayotte, France.
- Sana Chaabane Khelifi, Multilevel Methods for Computational Fluid Dynamics (2009-2013) - Died of a long illness.
- Thu Huyen Dao, Domain Decomposition Methods for Computational Fluid Dynamics (2009-2013) - Assistant Professor at the Hanoi University of Agriculture, Vietnam.
- Thomas Cadeau, Parallel in Time Domain Decomposition Methods for Electricity Network (2009-2012) - Development Engineer at Intitek, France.
- Florent Pruvost, Parallel and Distributed Algorithms for Electricity Network (2008-2012) - Research and Development Engineer at Université Bordeaux 1, France.
- Fei Teng, Economic-based Scheduling Algorithms in Cloud Computing (2008-2011) - Assistant Professor at Southwest Jiaotong University, China.
- Haixiang Zhao, Dynamic Data Mining Algorithms for Energy Efficiency (2008-2011) - Research and Development Engineer at Amadeus S.A., France.
- Cédric Venet, Parallel and Distributed Fault Tolerant Algorithms for Acoustic Problems (2007-2011) - Research and Development Engineer at Schlumberger, France.
- Jie Pan, Dynamic Data Management in Distributed Heterogeneous Systems (2007-2010) - Development Engineer at Klee Group, France.
- Thi Mai Huong Nguyen, Dynamic Access of Virtual Data via Web Services Resource (2006-2008) - Research and Development Engineer at IBM, Vietnam.
- Lei Yu, Dynamic Deployment of Scientific Applications based on Web Services Resource (2006-2008) - Assistant Professor at Beihang University, China.

Post-Doctoral Researchers Supervised

- Ahmed El Kerim (2023-2024) - Research Engineer at Safran Tech, France.
- Mengchen Wang (2022) - Research Engineer at MTB by Créative (MTB111), France.
- Qinneng Zou (2019-2020) - Assistant Professor at Beijing University of Posts and Telecommunications, China.
- Guillaume Benissan (2017-2018) - Assistant Professor at Peoples' Friendship University of Russia, Russia.
- Hanyu Zhang (2016-2017) - Assistant Professor at Providence University, Taiwan.
- Abal-Kassim Cheik Ahamed (2015-2016) - Assistant Professor at Université de Mayotte, France.
- Atsushi Suzuki (2013-2014) - Senior Research Engineer, Kyoto University, Japan.
- Takao Tsuji (2013-2014) - Assistant Professor, Yokohama National University, Japan.
- Jeonghoon Lee (2011-2012) - Assistant Professor, Korea Advanced Institute of Science and Technology Daejeon, Republic of Korea.
- Cédric Venet (2011) - Research and Development Engineer at Schlumberger, France.
- Somanchi K Murthy (2010-2011) - Assistant Professor, Indian Institute of Technology Kanpur, India.

Research Assistants Supervised

- Ricardo Bispo Vieira (2014) - Université de Versailles Saint Quentin en Yvelines, France.
- Rémi Cerise (2012-2013) - Institut National Polytechnique de Grenoble, France.
- Guillaume Gbikpi-Benissan (2012-2013) - Université Pierre et Marie Curie, France.
- Thi Phuong Kieu Nguyen (2011-2012) - Ho Chi Minh City University of Pedagogy, Vietnam.
- Alain Mugabe Rucyahana (2011-2012) - Université Catholique de Louvain, Belgium.
- Lve Zhang (2008-2009) - Harbin Institute of Technology, China.
- Qinyuan Deng (2007-2008) - École Normale Supérieure de Cachan, France.
- Qingfeng Fan (2007-2008) - Wuhan University, China.

Master's Students Supervised

Supervision of over 300 Master's theses (2006-2026), averaging 14-16 per year, in finite element analysis, partitioning techniques, domain decomposition methods, scientific machine learning, high-performance computing, audio signal processing and sound holography.

(Detailed list available upon request)

Université Henri Poincaré

Courses Taught

- Foundations of Pure Mathematics - Undergraduate level.

Lecture Notes and Online Tutorials

- [1] C.-H. Lai and F. Magoulès. Decomposition of fields in aeroacoustics. University of Greenwich, Greenwich, United Kingdom. E-learning: 30 hours, Sep 2004.
- [2] C.-H. Lai and F. Magoulès. Decomposition of domains in acoustics. University of Greenwich, Greenwich, United Kingdom. E-learning: 30 hours, Sep 2004.
- [3] L. Rosier and F. Magoulès. Fonctions de plusieurs variables et formes différentielles. Université Henri Poincaré, Nancy, France. Paperback: 82 pages, E-learning: 30 hours, May 2003.
- [4] L. Rosier and F. Magoulès. Topologie dans R^N . Université Henri Poincaré, Nancy, France. Paperback: 70 pages, E-learning: 30 hours, May 2003.
- [5] B. Topping, J. Muylle, P. Ivanyi, B. Cheng, R. Putanowicz, F. Magoulès, P.-A. Gueneguo, S. Muller, and D. Roegel. Mesh generation methods. Heriot-Watt University, Edinburgh, United Kingdom. E-learning: 30 hours, Jan 2003.
- [6] F. Magoulès. Initiation à la programmation en Tcl. Université Henri Poincaré, Nancy, France. Paperback: 46 pages, E-learning: 30 hours, Nov 2002.

Master Students Supervised

Supervision of ten Master's theses (2000-2006), averaging 1-2 per year, in finite element analysis and related data, interactive visualization, audio signal processing, and motion capture.

(Detailed list available upon request)

Undergraduate Students Supervised

Supervision of four undergraduate research projects (2000-2006) in partitioning techniques, domain decomposition methods, and large-scale databases.

(Detailed list available upon request)

Université Pierre et Marie Curie

Courses Taught

- Numerical Linear Algebra - Graduate level.
- Introduction to Scilab - Graduate level.
- Linear Algebra - Undergraduate level.

RESEARCH GRANTS AND CONTRACTS (FUNDED)

Principal Investigator or Co-Principal Investigator

Actively participated in research funding activities, serving as principal investigator or co-investigator for national and international grants supporting projects in numerical analysis, scientific computing, high performance computing, and scientific machine learning.

- Programme Investissement d'Avenir (PIA), URBA(IA), un jumeau numérique au service de l'urbanisme (2024-2028).
- Agence Nationale de la Recherche, Localisation des Ondes pour la Reduction des Emissions Acoustiques Large Bande, (2024-2028).
- Transvalor SA., Academic Chair of Artificial Intelligence for Finite Element Methods, (2022-2026).
- Transvalor SA., Artificial Intelligence Approaches for Enhancing Finite Element Methods (2021-2022).
- Equipement d'Excellence (EQUIPEX), Continuum: Un réseau de plateformes pour la visualisation interactive (SINAPSE), (2020-present).
- Fondation CentraleSupélec, Réalisation d'un jumeau numérique pour le traitement des acouphènes, (2021-2022).
- Université Paris-Saclay, Département STIC, Analyse de Posture pour Usager de Fauteuil Roulant, (2019).
- Agence Nationale de la Recherche, Asynchronous Domain Decomposition Methods, (2018-2023).
- ECOJOKO, Artificial Intelligence for Electricity Savings in Buildings, (2018-2021).
- CNRS (INSU), Nouvelles Approches de Parallélisation pour l'Océan et l'Atmosphère: la dimension temporelle (AO-LEFE), (2016-2017).
- Institut de Recherche Technologique (IRT SystemX), Algorithmes Parallèles (APA), (2014-2017).
- Laboratoire International Associé (LIA2MCIS), Power Aware Scheduling of Real-time data processing in Big Data Centre, (2013).
- Equipement d'Excellence (EQUIPEX), Digiscope: Un réseau de plateformes pour la visualisation interactive (SINAPSE), (2011-2019).
- European Commission, Collaborative Research into Exascale Systemware, Tools and Applications (CRESTA), (2011-2014).
- Laboratoire International Associé (LIA2MCIS), Fault Tolerance for the Self Healing Overlay Network (SHON) middleware, (2011, 2012).
- Pôle de Compétitivité Cap Digital, Simulations architecturales réalistes et immersives (Callisto), (2010-2013).
- Pôle de Compétitivité Systematic, Plateforme intégrée de parallélisation de codes industriels pour architectures GPU & hybrides (OpenGPU), (2010-2012).
- Laboratoire franco-chinois de recherche en informatique, automatique et mathématiques appliquées (LIAMA), Large Scale Parallel and Distributed Systems, (2010).
- Electricité De France, Algebraic Multigrid Methods Applied to Convection-Diffusion, (2010-2013).
- European Commission, Pan European Grid Advanced Simulation and state Estimation (PEGASE), (2008-2012).
- SAP, Academic Chair of Business Intelligence, (2008-2013).
- Business Objects, Replicated Data Management in Distributed Systems, (2008-2011).

- Pôle de Recherche et d'Enseignement Supérieur UniverSud Paris, Assistance ambiante pour le handicap, (2007-2008).
- Projet PluriFormation (PPF), École Centrale Paris, Cloud Computing for Econometrics, (2007-2010).
- Pôle de Compétitivité Cap Digital, Territoire augmenté (Terra Magna), (2007-2009).
- Pôle de Compétitivité Systematic, PetaOpérations Par Seconde (POPS), (2007-2009).
- Business Objects, Data Management in Distributed Heterogeneous Systems, (2007-2010).
- Terra Nova S.A., Artificial Intelligence in District Heating Systems, (2007-2008).
- European Commission, Enabling Grids for E-science (EGEE), (2006-2008).
- European Commission, Business Experiments in Grid (BeInGrid), (2006-2008).
- Pôle de Compétitivité Systematic, Une nouvelle génération de serveurs pour le calcul intensif et le traitement de l'information (FAME2), (2005-2007).
- Bonus Qualité Recherche Internationale, Université Henri Poincaré, Development of Mathematical and Scientific Computing Tools for Earthquake Seismology Problems, (2006).
- Ministère des Affaires Etrangères and British Council, Parallel Algorithms for Aero-Acoustics Problems, (2004 & 2005).
- Bonus Qualité Recherche Internationale, Université Henri Poincaré, Development of Mathematical and Scientific Computing Tools for Engineering Problems, (2003).
- Ministère des Affaires Etrangères and British Council, Parallel and Distributed Computing for Engineering Problems, (2002 & 2004).
- Hutchinson S.A., Development of Visualization Tools for Engineering Problems, (2002).
- European Commission, Domain Decomposition Methods for Integrated Noise and Structural Analysis (DOMINOS), (1997-2000).

PUBLICATIONS

More than 300 publications, including more than 100 peer-reviewed journal articles. [See Complete Publications List (PDF)]

1st October 2026